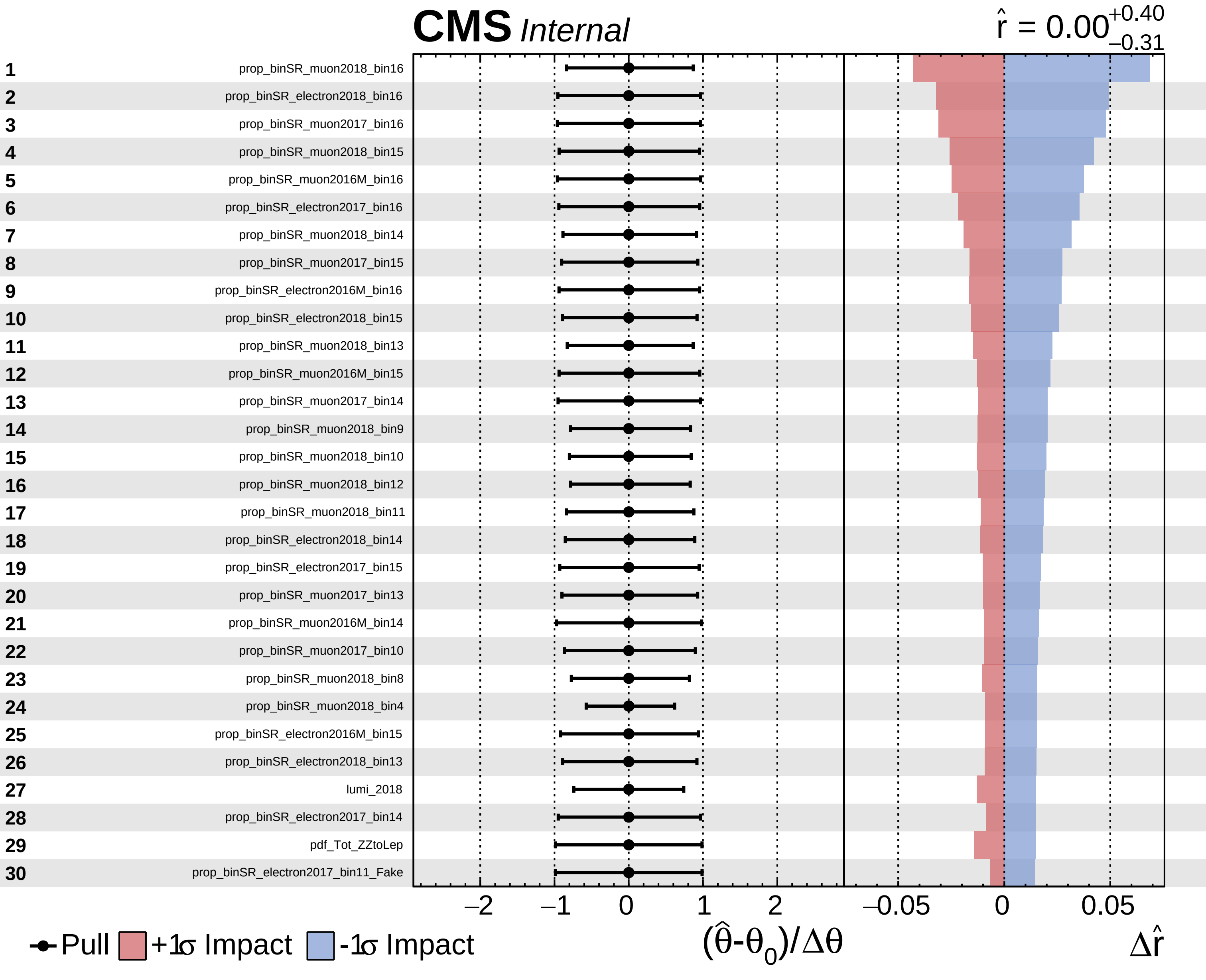
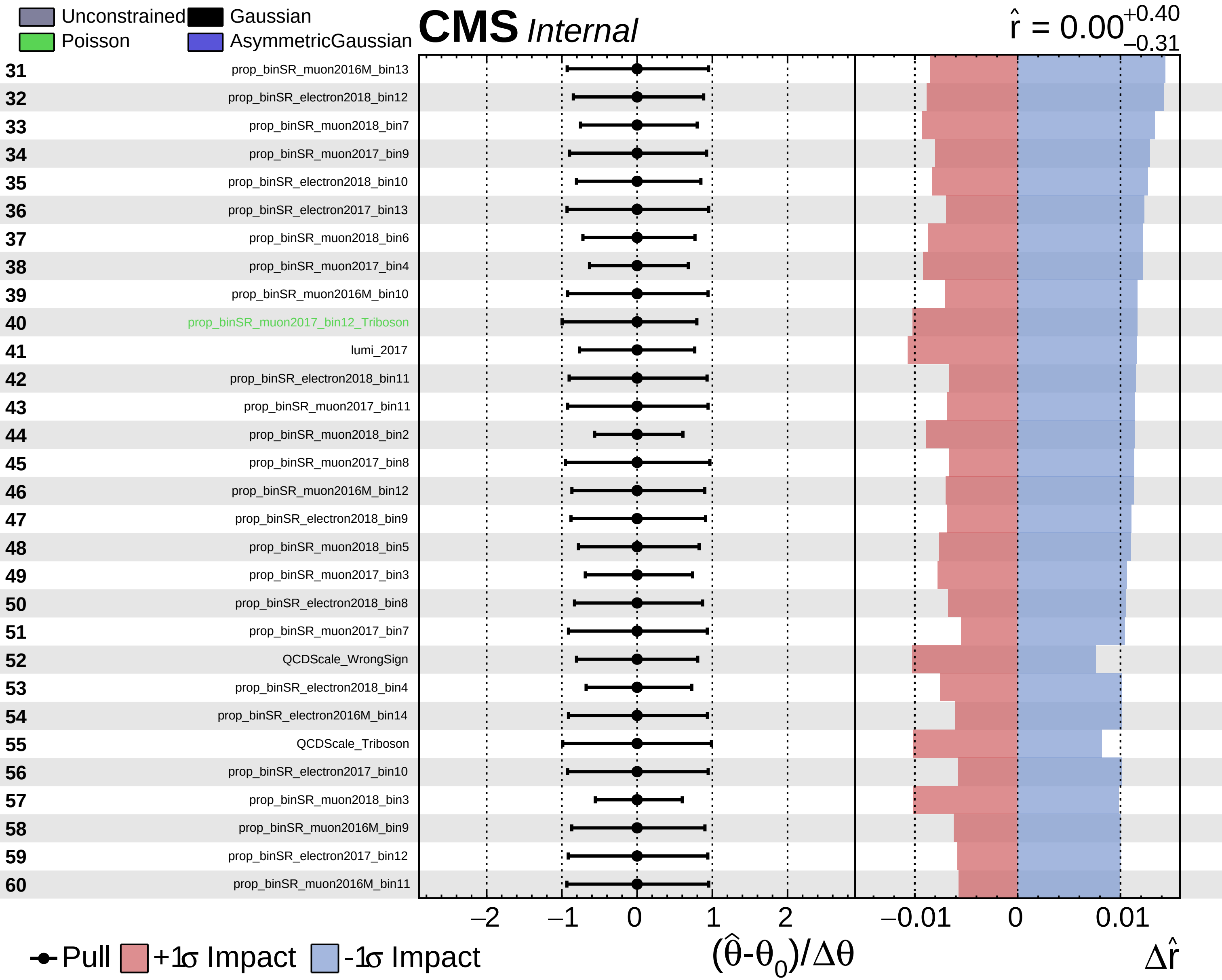
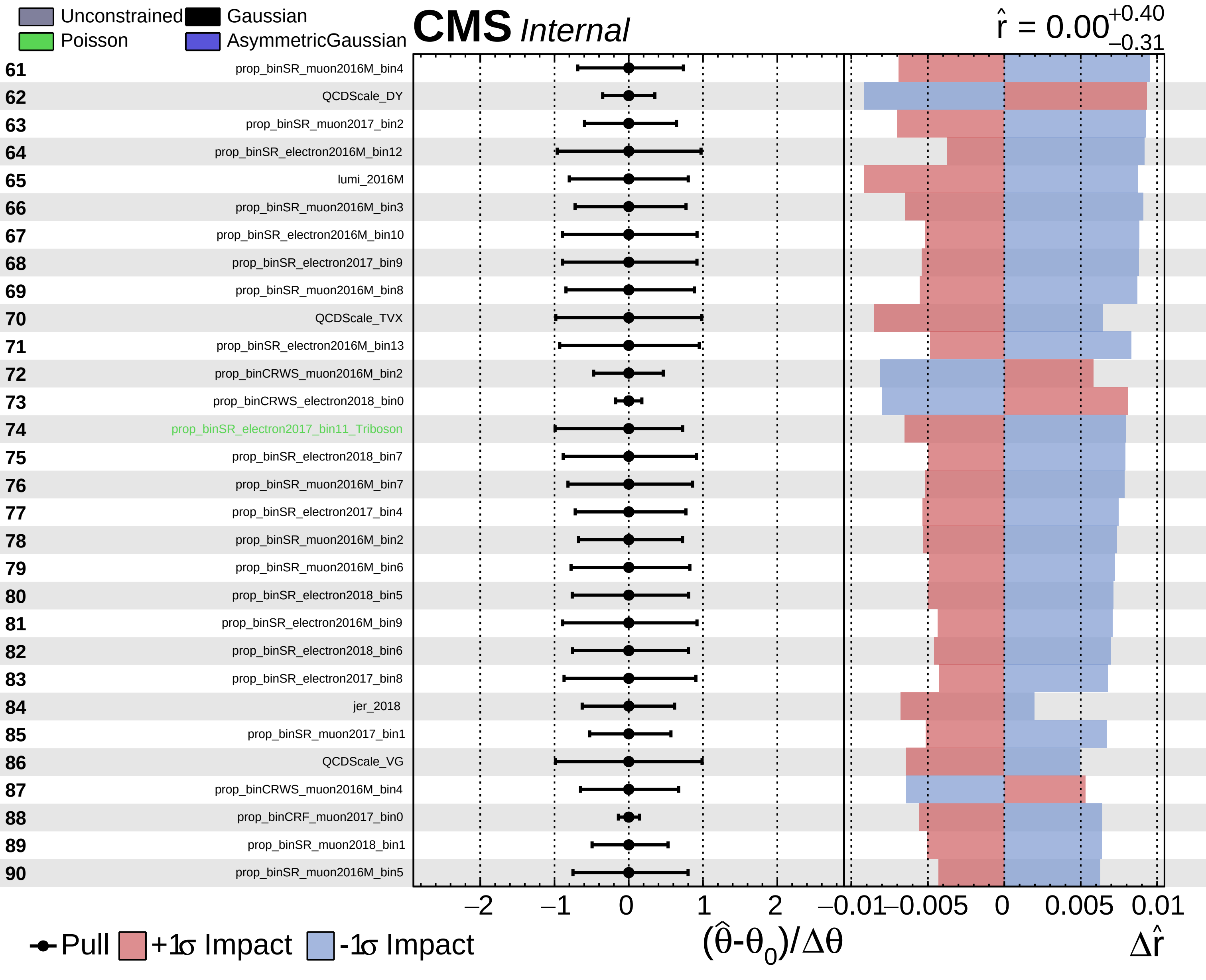
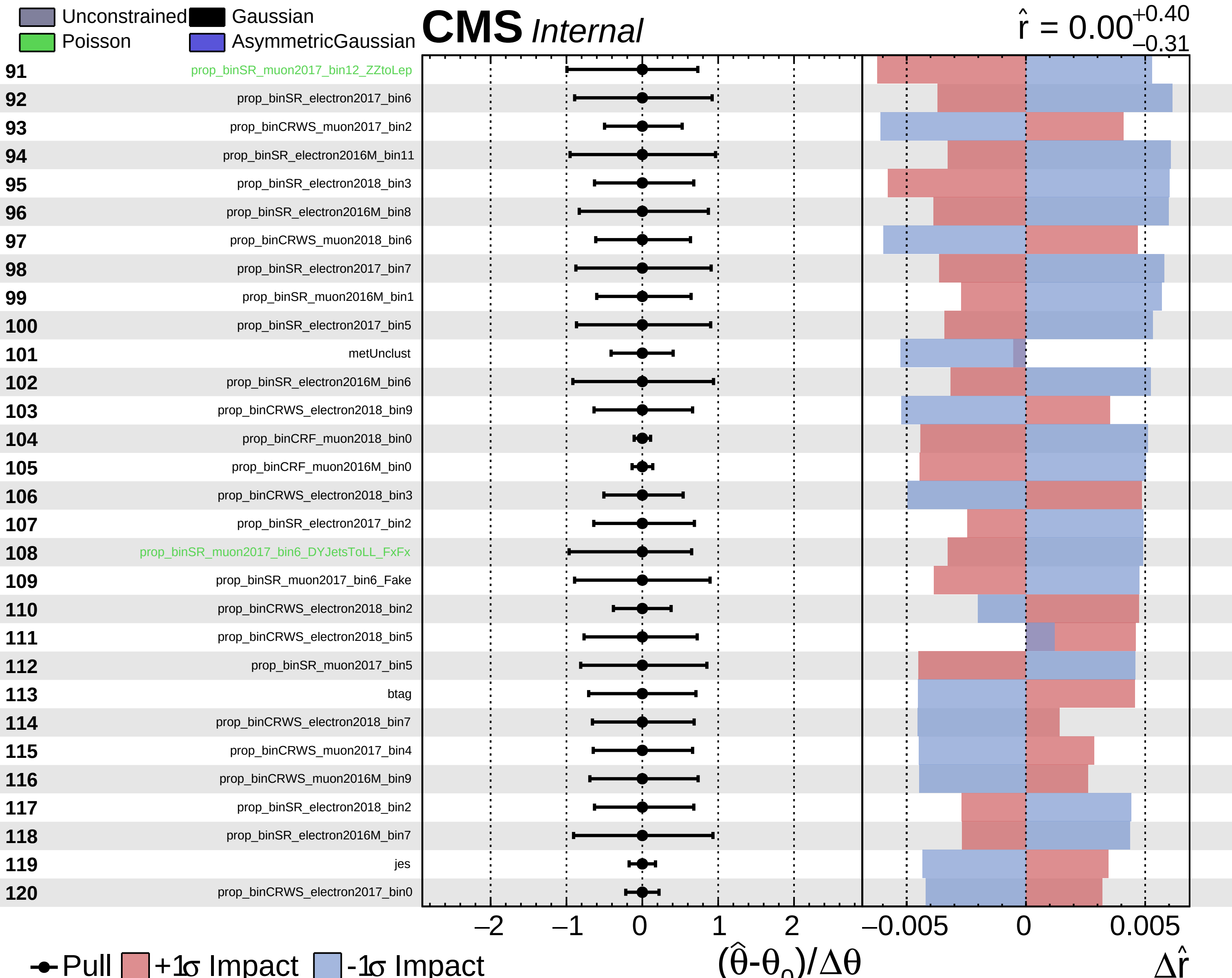

CMS Internal
r̂ = 0.00 +0.40 −0.31
1 prop_binSR_muon2018_bin16
2 prop_binSR_electron2018_bin16
3 prop_binSR_muon2017_bin16
4 prop_binSR_muon2018_bin15
5 prop_binSR_muon2016M_bin16
6 prop_binSR_electron2017_bin16
7 prop_binSR_muon2018_bin14
8 prop_binSR_muon2017_bin15
9 prop_binSR_electron2016M_bin16
10 prop_binSR_electron2018_bin15
11 prop_binSR_muon2018_bin13
12 prop_binSR_muon2016M_bin15
13 prop_binSR_muon2017_bin14
14 prop_binSR_muon2018_bin9
15 prop_binSR_muon2018_bin10
16 prop_binSR_muon2018_bin12
17 prop_binSR_muon2018_bin11
18 prop_binSR_electron2018_bin14
19 prop_binSR_electron2017_bin15
20 prop_binSR_muon2017_bin13
21 prop_binSR_muon2016M_bin14
22 prop_binSR_muon2017_bin10
23 prop_binSR_muon2018_bin8
24 prop_binSR_muon2018_bin4
25 prop_binSR_electron2016M_bin15
26 prop_binSR_electron2018_bin13
27 lumi_2018
28 prop_binSR_electron2017_bin14
29 pdf_Tot_ZZtoLep
30 prop_binSR_electron2017_bin11_Fake
−2 −1 0 1 2
−0.05 0 0.05
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂



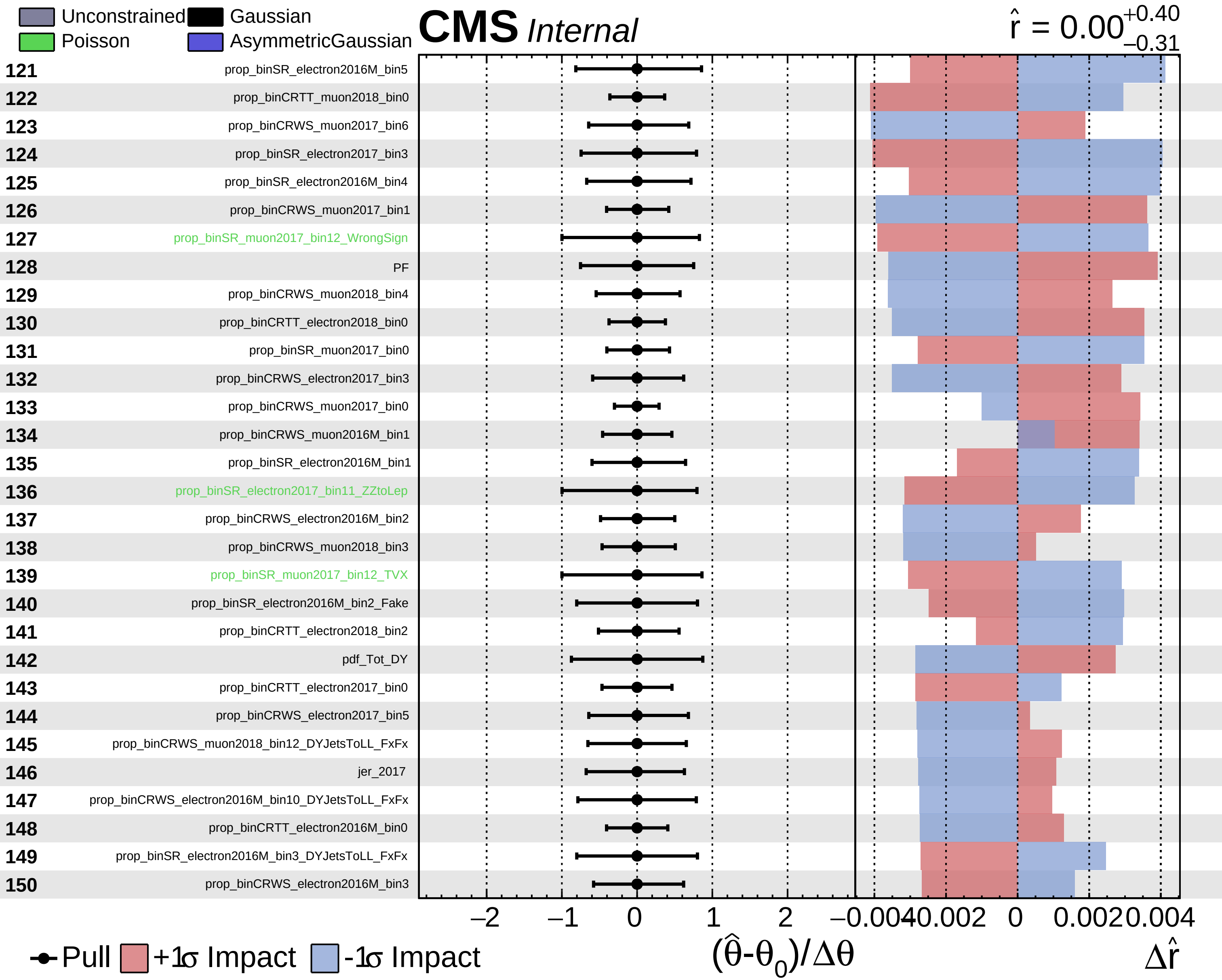
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
31 prop_binSR_muon2016M_bin13
32 prop_binSR_electron2018_bin12
33 prop_binSR_muon2018_bin7
34 prop_binSR_muon2017_bin9
35 prop_binSR_electron2018_bin10
36 prop_binSR_electron2017_bin13
37 prop_binSR_muon2018_bin6
38 prop_binSR_muon2017_bin4
39 prop_binSR_muon2016M_bin10
40 prop_binSR_muon2017_bin12_Triboson
41 lumi_2017
42 prop_binSR_electron2018_bin11
43 prop_binSR_muon2017_bin11
44 prop_binSR_muon2018_bin2
45 prop_binSR_muon2017_bin8
46 prop_binSR_muon2016M_bin12
47 prop_binSR_electron2018_bin9
48 prop_binSR_muon2018_bin5
49 prop_binSR_muon2017_bin3
50 prop_binSR_electron2018_bin8
51 prop_binSR_muon2017_bin7
52 QCDScale_WrongSign
53 prop_binSR_electron2018_bin4
54 prop_binSR_electron2016M_bin14
55 QCDScale_Triboson
56 prop_binSR_electron2017_bin10
57 prop_binSR_muon2018_bin3
58 prop_binSR_muon2016M_bin9
59 prop_binSR_electron2017_bin12
60 prop_binSR_muon2016M_bin11
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



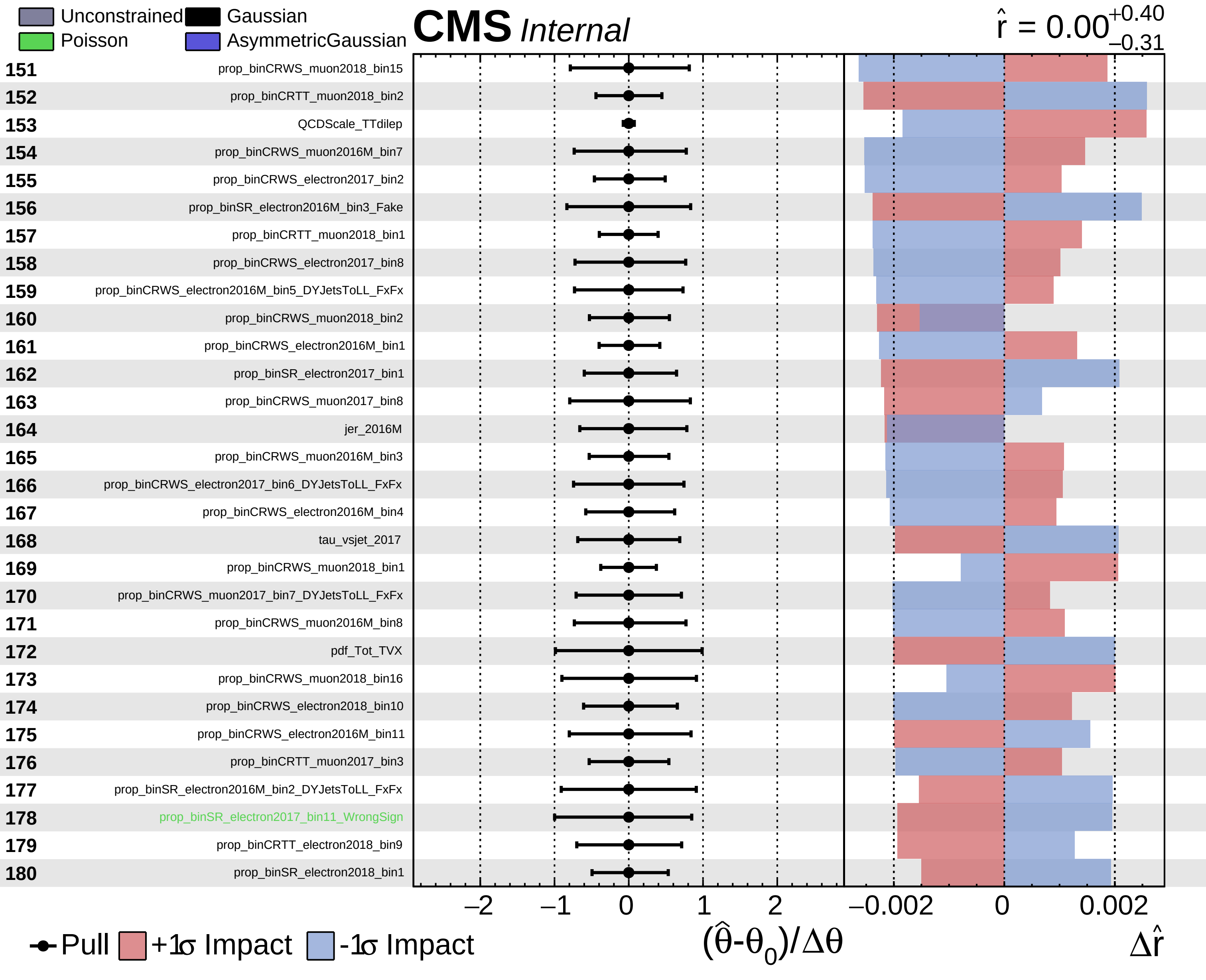
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
61 prop_binSR_muon2016M_bin4
62 QCDScale_DY
63 prop_binSR_muon2017_bin2
64 prop_binSR_electron2016M_bin12
65 lumi_2016M
66 prop_binSR_muon2016M_bin3
67 prop_binSR_electron2016M_bin10
68 prop_binSR_electron2017_bin9
69 prop_binSR_muon2016M_bin8
70 QCDScale_TVX
71 prop_binSR_electron2016M_bin13
72 prop_binCRWS_muon2016M_bin2
73 prop_binCRWS_electron2018_bin0
74 prop_binSR_electron2017_bin11_Triboson
75 prop_binSR_electron2018_bin7
76 prop_binSR_muon2016M_bin7
77 prop_binSR_electron2017_bin4
78 prop_binSR_muon2016M_bin2
79 prop_binSR_muon2016M_bin6
80 prop_binSR_electron2018_bin5
81 prop_binSR_electron2016M_bin9
82 prop_binSR_electron2018_bin6
83 prop_binSR_electron2017_bin8
84 jer_2018
85 prop_binSR_muon2017_bin1
86 QCDScale_VG
87 prop_binCRWS_muon2016M_bin4
88 prop_binCRF_muon2017_bin0
89 prop_binSR_muon2018_bin1
90 prop_binSR_muon2016M_bin5
−2
−1
0
1
2
−0.01
−0.005
0
0.005
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



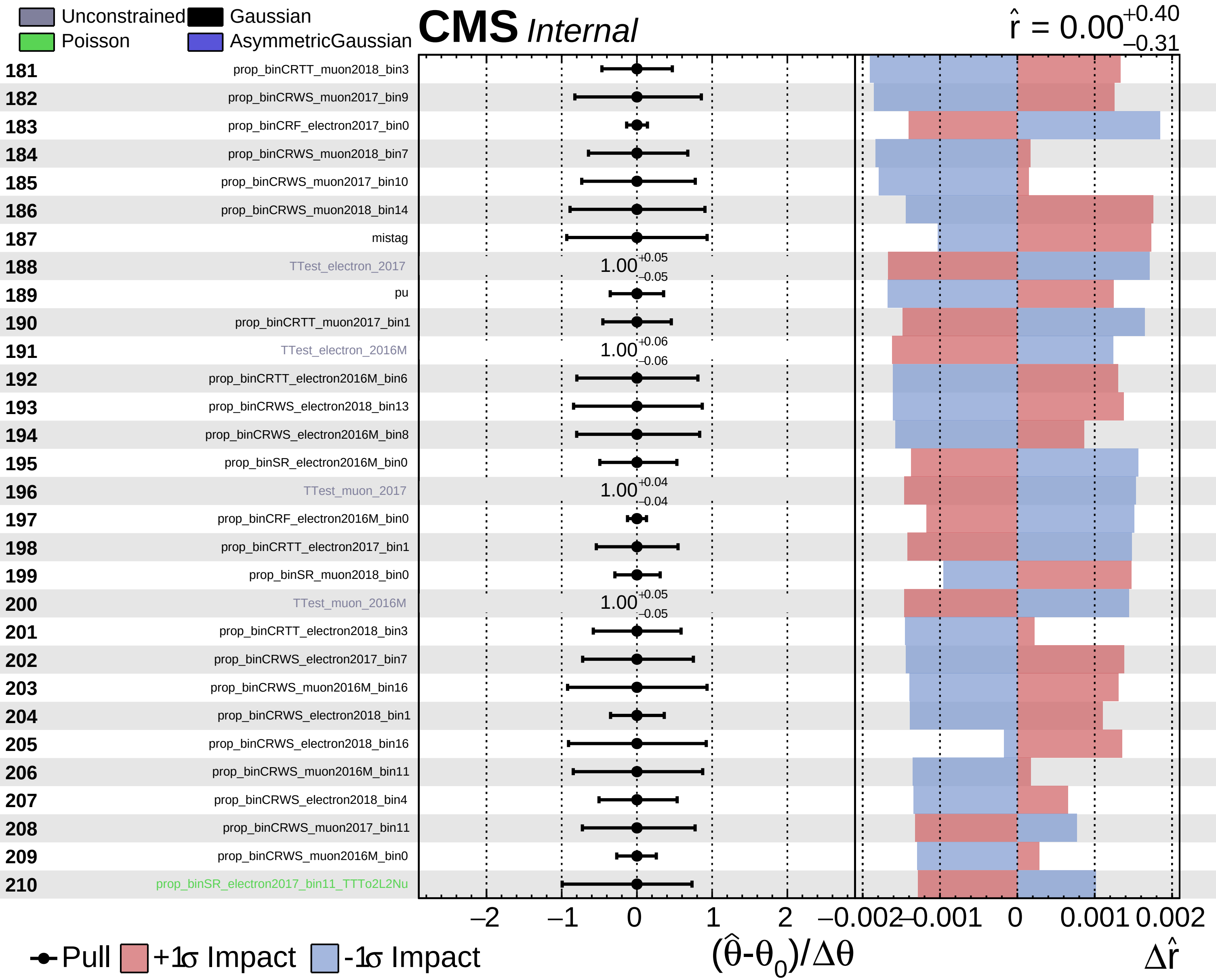
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.40 −0.31
91 prop_binSR_muon2017_bin12_ZZtoLep
92 prop_binSR_electron2017_bin6
93 prop_binCRWS_muon2017_bin2
94 prop_binSR_electron2016M_bin11
95 prop_binSR_electron2018_bin3
96 prop_binSR_electron2016M_bin8
97 prop_binCRWS_muon2018_bin6
98 prop_binSR_electron2017_bin7
99 prop_binSR_muon2016M_bin1
100 prop_binSR_electron2017_bin5
101 metUnclust
102 prop_binSR_electron2016M_bin6
103 prop_binCRWS_electron2018_bin9
104 prop_binCRF_muon2018_bin0
105 prop_binCRF_muon2016M_bin0
106 prop_binCRWS_electron2018_bin3
107 prop_binSR_electron2017_bin2
108 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
109 prop_binSR_muon2017_bin6_Fake
110 prop_binCRWS_electron2018_bin2
111 prop_binCRWS_electron2018_bin5
112 prop_binSR_muon2017_bin5
113 btag
114 prop_binCRWS_electron2018_bin7
115 prop_binCRWS_muon2017_bin4
116 prop_binCRWS_muon2016M_bin9
117 prop_binSR_electron2018_bin2
118 prop_binSR_electron2016M_bin7
119 jes
120 prop_binCRWS_electron2017_bin0
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



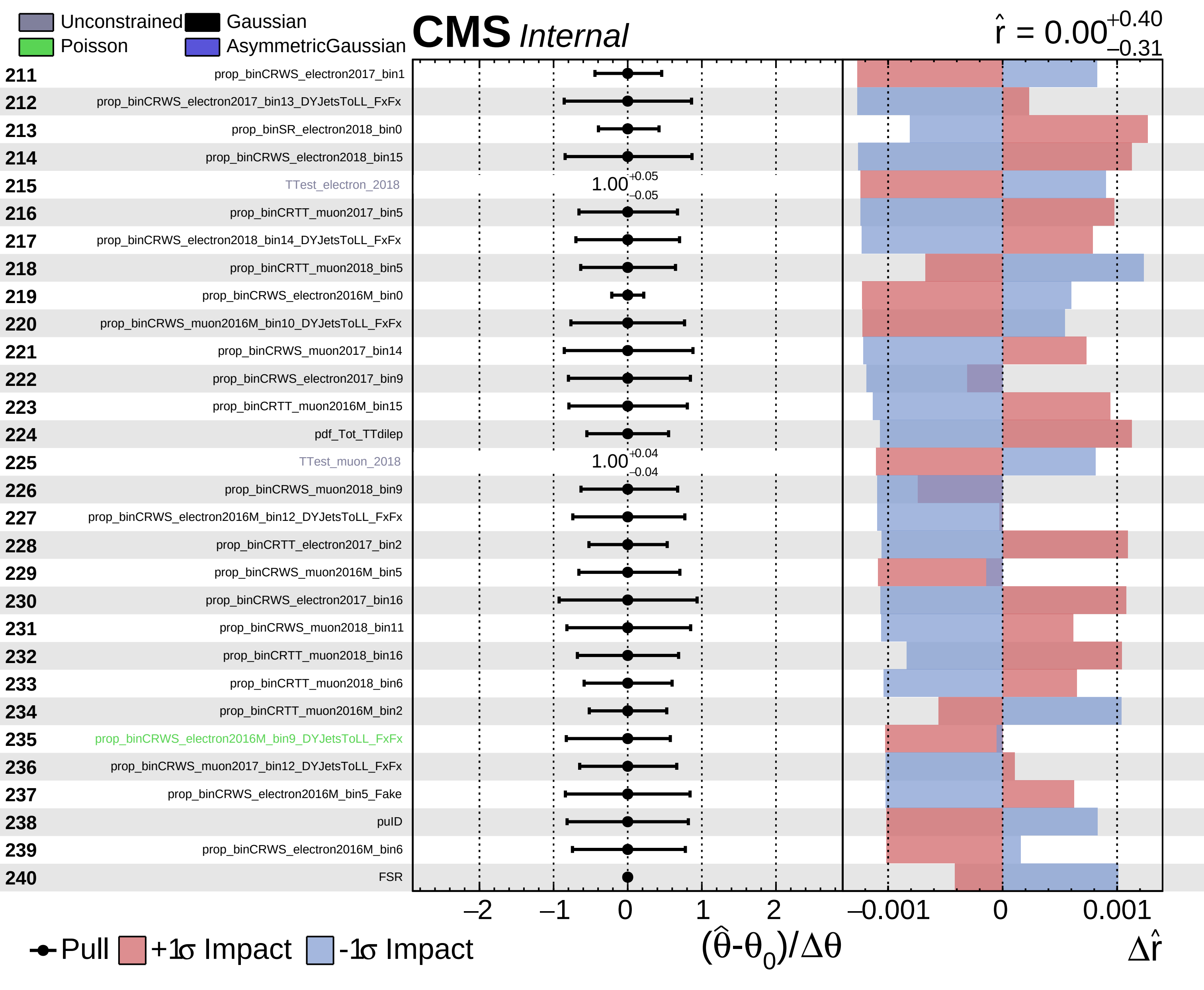
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.40 −0.31
121 prop_binSR_electron2016M_bin5
122 prop_binCRTT_muon2018_bin0
123 prop_binCRWS_muon2017_bin6
124 prop_binSR_electron2017_bin3
125 prop_binSR_electron2016M_bin4
126 prop_binCRWS_muon2017_bin1
127 prop_binSR_muon2017_bin12_WrongSign
128 PF
129 prop_binCRWS_muon2018_bin4
130 prop_binCRTT_electron2018_bin0
131 prop_binSR_muon2017_bin0
132 prop_binCRWS_electron2017_bin3
133 prop_binCRWS_muon2017_bin0
134 prop_binCRWS_muon2016M_bin1
135 prop_binSR_electron2016M_bin1
136 prop_binSR_electron2017_bin11_ZZtoLep
137 prop_binCRWS_electron2016M_bin2
138 prop_binCRWS_muon2018_bin3
139 prop_binSR_muon2017_bin12_TVX
140 prop_binSR_electron2016M_bin2_Fake
141 prop_binCRTT_electron2018_bin2
142 pdf_Tot_DY
143 prop_binCRTT_electron2017_bin0
144 prop_binCRWS_electron2017_bin5
145 prop_binCRWS_muon2018_bin12_DYJetsToLL_FxFx
146 jer_2017
147 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
148 prop_binCRTT_electron2016M_bin0
149 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
150 prop_binCRWS_electron2016M_bin3
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂



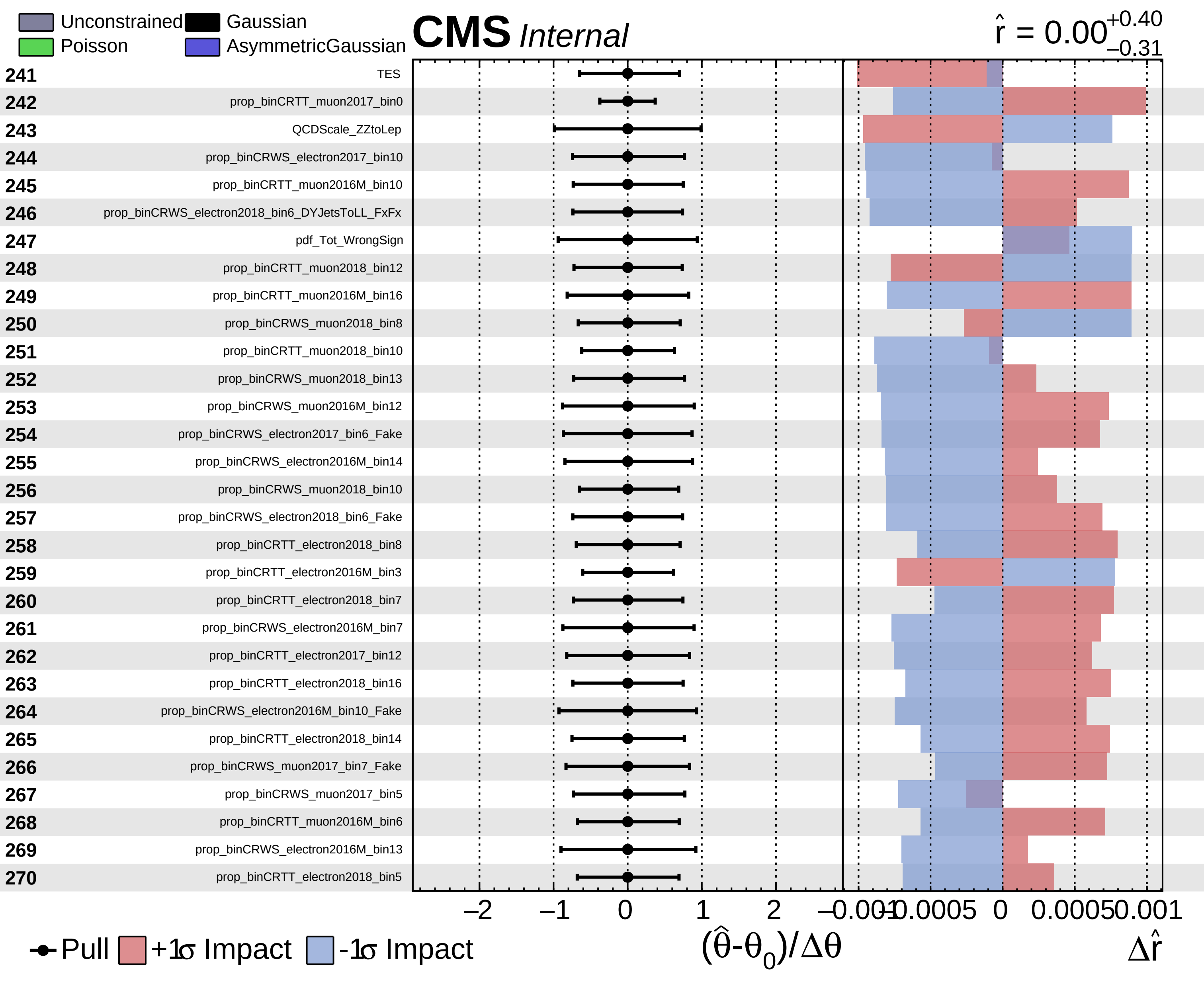
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
151 prop_binCRWS_muon2018_bin15
152 prop_binCRTT_muon2018_bin2
153 QCDScale_TTdilep
154 prop_binCRWS_muon2016M_bin7
155 prop_binCRWS_electron2017_bin2
156 prop_binSR_electron2016M_bin3_Fake
157 prop_binCRTT_muon2018_bin1
158 prop_binCRWS_electron2017_bin8
159 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
160 prop_binCRWS_muon2018_bin2
161 prop_binCRWS_electron2016M_bin1
162 prop_binSR_electron2017_bin1
163 prop_binCRWS_muon2017_bin8
164 jer_2016M
165 prop_binCRWS_muon2016M_bin3
166 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
167 prop_binCRWS_electron2016M_bin4
168 tau_vsjet_2017
169 prop_binCRWS_muon2018_bin1
170 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
171 prop_binCRWS_muon2016M_bin8
172 pdf_Tot_TVX
173 prop_binCRWS_muon2018_bin16
174 prop_binCRWS_electron2018_bin10
175 prop_binCRWS_electron2016M_bin11
176 prop_binCRTT_muon2017_bin3
177 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
178 prop_binSR_electron2017_bin11_WrongSign
179 prop_binCRTT_electron2018_bin9
180 prop_binSR_electron2018_bin1
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



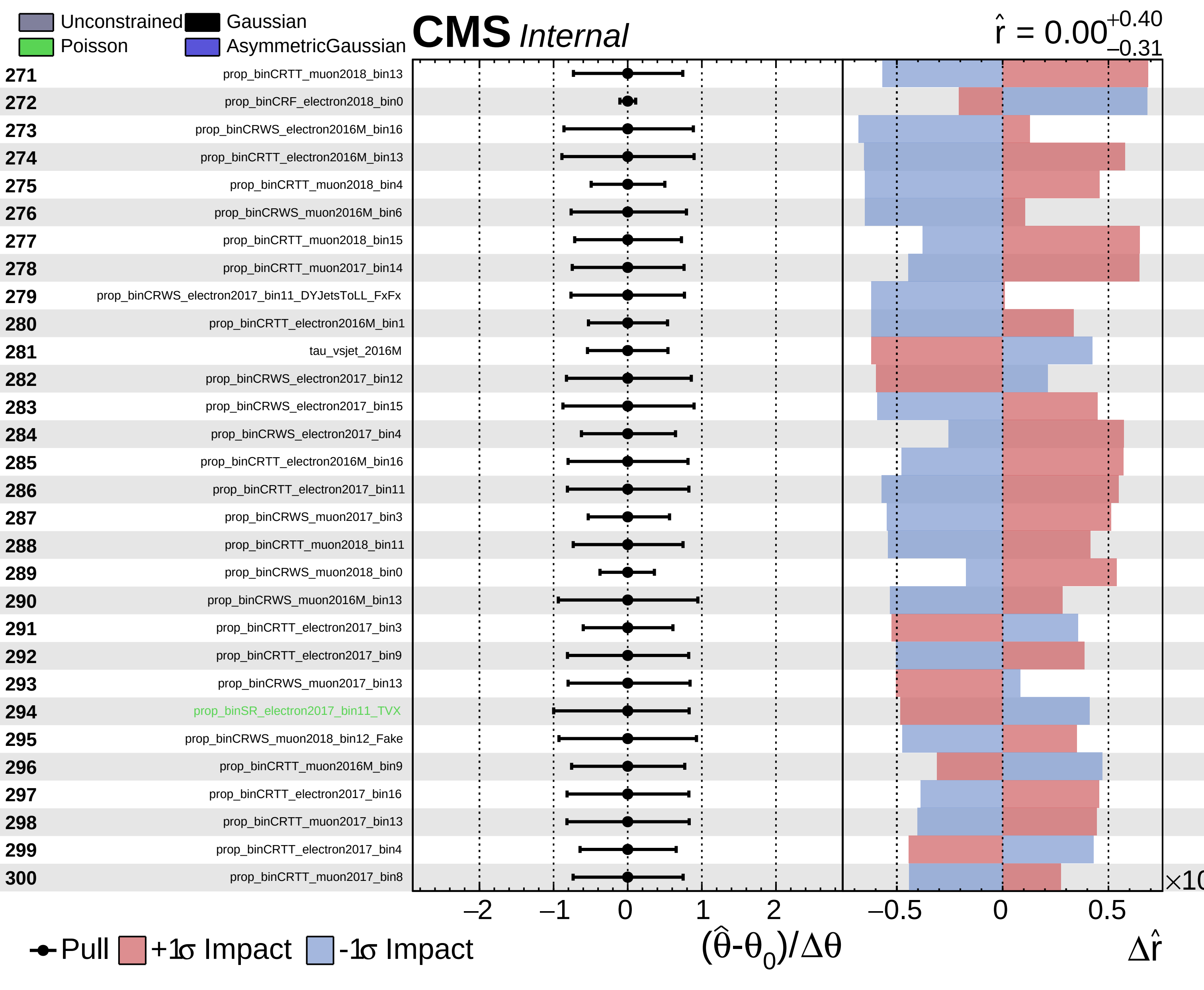
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.40 −0.31
181 prop_binCRTT_muon2018_bin3
182 prop_binCRWS_muon2017_bin9
183 prop_binCRF_electron2017_bin0
184 prop_binCRWS_muon2018_bin7
185 prop_binCRWS_muon2017_bin10
186 prop_binCRWS_muon2018_bin14
187 mistag
188 TTest_electron_2017
1.00 +0.05 −0.05
189 pu
190 prop_binCRTT_muon2017_bin1
191 TTest_electron_2016M
1.00 +0.06 −0.06
192 prop_binCRTT_electron2016M_bin6
193 prop_binCRWS_electron2018_bin13
194 prop_binCRWS_electron2016M_bin8
195 prop_binSR_electron2016M_bin0
196 TTest_muon_2017
1.00 +0.04 −0.04
197 prop_binCRF_electron2016M_bin0
198 prop_binCRTT_electron2017_bin1
199 prop_binSR_muon2018_bin0
200 TTest_muon_2016M
1.00 +0.05 −0.05
201 prop_binCRTT_electron2018_bin3
202 prop_binCRWS_electron2017_bin7
203 prop_binCRWS_muon2016M_bin16
204 prop_binCRWS_electron2018_bin1
205 prop_binCRWS_electron2018_bin16
206 prop_binCRWS_muon2016M_bin11
207 prop_binCRWS_electron2018_bin4
208 prop_binCRWS_muon2017_bin11
209 prop_binCRWS_muon2016M_bin0
210 prop_binSR_electron2017_bin11_TTTo2L2Nu
−2
−1
0
1
2
−0.002
−0.001
0
0.001
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

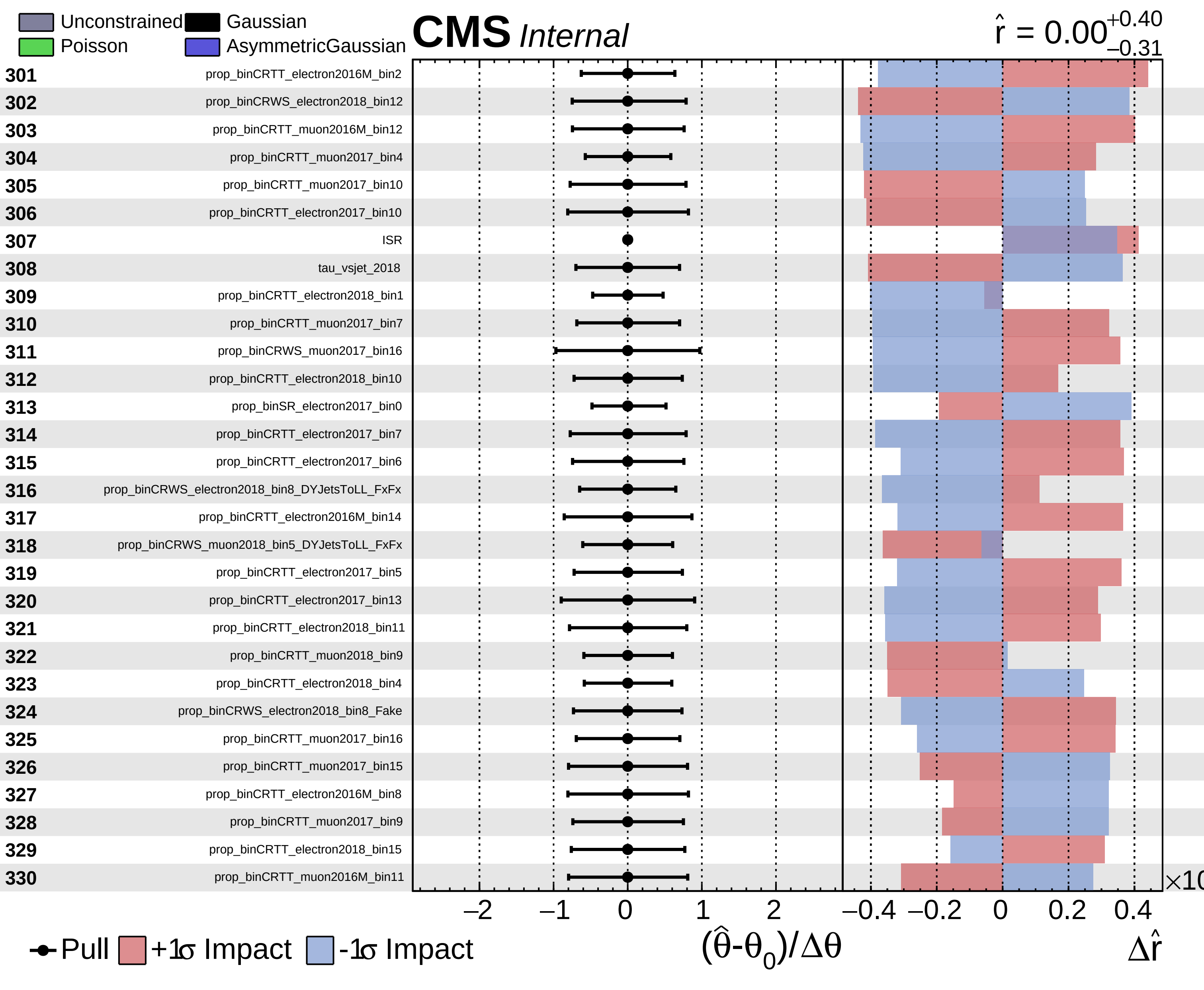


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
211 prop_binCRWS_electron2017_bin1
212 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
213 prop_binSR_electron2018_bin0
214 prop_binCRWS_electron2018_bin15
215 TTest_electron_2018 $1.00^{+0.05}_{-0.05}$
216 prop_binCRTT_muon2017_bin5
217 prop_binCRWS_electron2018_bin14_DYJetsToLL_FxFx
218 prop_binCRTT_muon2018_bin5
219 prop_binCRWS_electron2016M_bin0
220 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
221 prop_binCRWS_muon2017_bin14
222 prop_binCRWS_electron2017_bin9
223 prop_binCRTT_muon2016M_bin15
224 pdf_Tot_TTdilep
225 TTest_muon_2018 $1.00^{+0.04}_{-0.04}$
226 prop_binCRWS_muon2018_bin9
227 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
228 prop_binCRTT_electron2017_bin2
229 prop_binCRWS_muon2016M_bin5
230 prop_binCRWS_electron2017_bin16
231 prop_binCRWS_muon2018_bin11
232 prop_binCRTT_muon2018_bin16
233 prop_binCRTT_muon2018_bin6
234 prop_binCRTT_muon2016M_bin2
235 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
236 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
237 prop_binCRWS_electron2016M_bin5_Fake
238 puID
239 prop_binCRWS_electron2016M_bin6
240 FSR
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

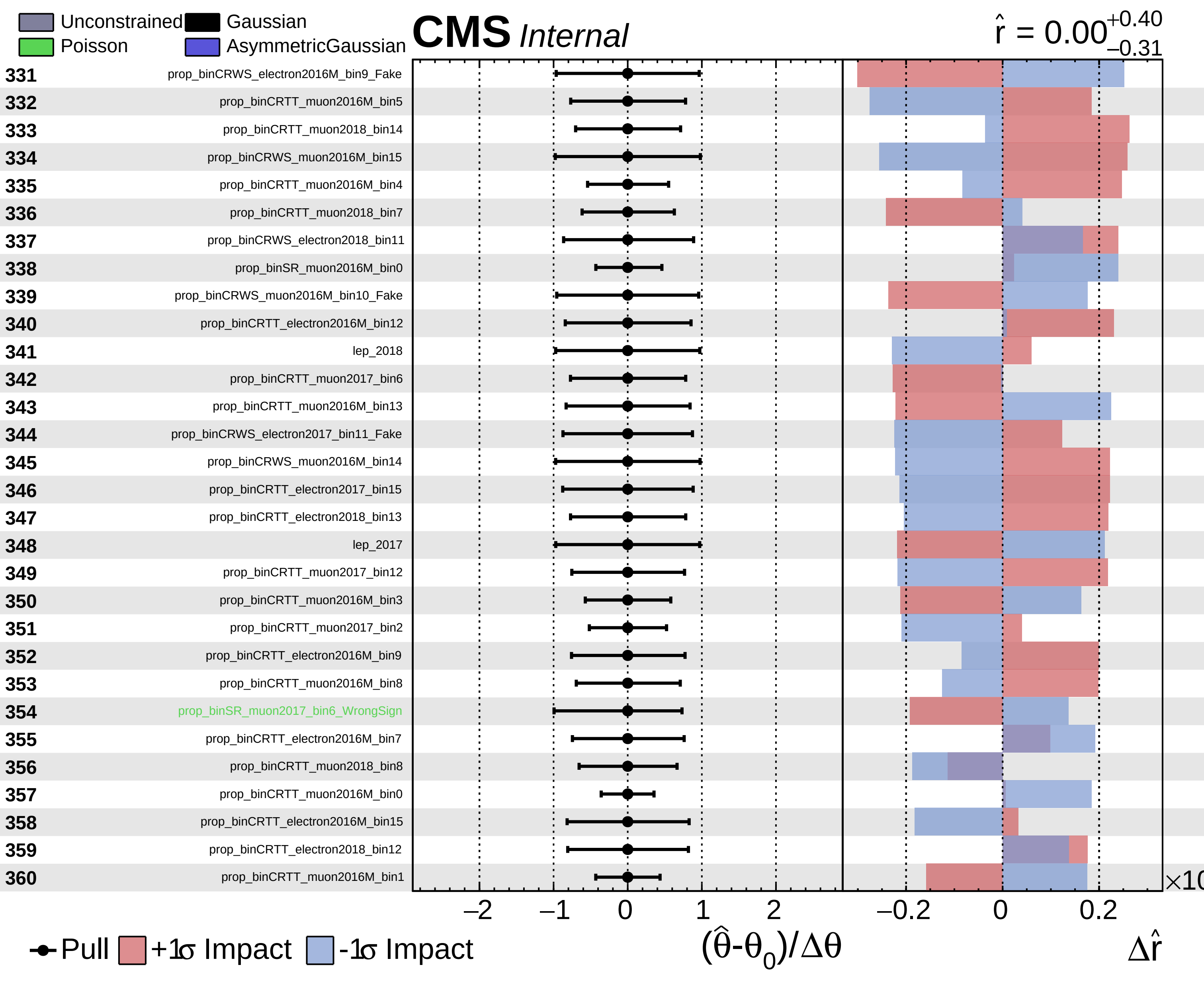


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
241 TES
242 prop_binCRTT_muon2017_bin0
243 QCDScale_ZZtoLep
244 prop_binCRWS_electron2017_bin10
245 prop_binCRTT_muon2016M_bin10
246 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
247 pdf_Tot_WrongSign
248 prop_binCRTT_muon2018_bin12
249 prop_binCRTT_muon2016M_bin16
250 prop_binCRWS_muon2018_bin8
251 prop_binCRTT_muon2018_bin10
252 prop_binCRWS_muon2018_bin13
253 prop_binCRWS_muon2016M_bin12
254 prop_binCRWS_electron2017_bin6_Fake
255 prop_binCRWS_electron2016M_bin14
256 prop_binCRWS_muon2018_bin10
257 prop_binCRWS_electron2018_bin6_Fake
258 prop_binCRTT_electron2018_bin8
259 prop_binCRTT_electron2016M_bin3
260 prop_binCRTT_electron2018_bin7
261 prop_binCRWS_electron2016M_bin7
262 prop_binCRTT_electron2017_bin12
263 prop_binCRTT_electron2018_bin16
264 prop_binCRWS_electron2016M_bin10_Fake
265 prop_binCRTT_electron2018_bin14
266 prop_binCRWS_muon2017_bin7_Fake
267 prop_binCRWS_muon2017_bin5
268 prop_binCRTT_muon2016M_bin6
269 prop_binCRWS_electron2016M_bin13
270 prop_binCRTT_electron2018_bin5
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

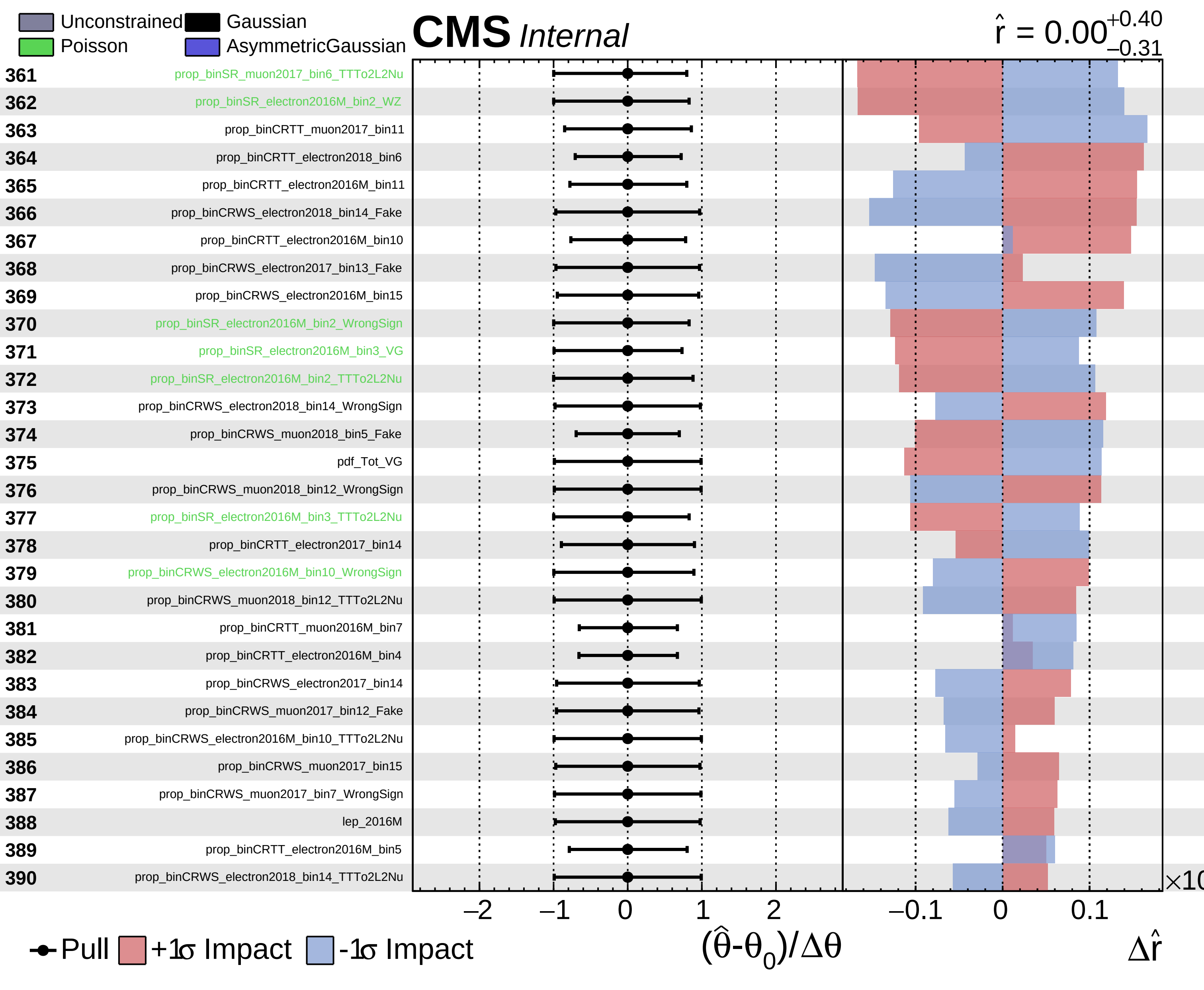

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.40 −0.31
271 prop_binCRTT_muon2018_bin13
272 prop_binCRF_electron2018_bin0
273 prop_binCRWS_electron2016M_bin16
274 prop_binCRTT_electron2016M_bin13
275 prop_binCRTT_muon2018_bin4
276 prop_binCRWS_muon2016M_bin6
277 prop_binCRTT_muon2018_bin15
278 prop_binCRTT_muon2017_bin14
279 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
280 prop_binCRTT_electron2016M_bin1
281 tau_vsjet_2016M
282 prop_binCRWS_electron2017_bin12
283 prop_binCRWS_electron2017_bin15
284 prop_binCRWS_electron2017_bin4
285 prop_binCRTT_electron2016M_bin16
286 prop_binCRTT_electron2017_bin11
287 prop_binCRWS_muon2017_bin3
288 prop_binCRTT_muon2018_bin11
289 prop_binCRWS_muon2018_bin0
290 prop_binCRWS_muon2016M_bin13
291 prop_binCRTT_electron2017_bin3
292 prop_binCRTT_electron2017_bin9
293 prop_binCRWS_muon2017_bin13
294 prop_binSR_electron2017_bin11_TVX
295 prop_binCRWS_muon2018_bin12_Fake
296 prop_binCRTT_muon2016M_bin9
297 prop_binCRTT_electron2017_bin16
298 prop_binCRTT_muon2017_bin13
299 prop_binCRTT_electron2017_bin4
300 prop_binCRTT_muon2017_bin8
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

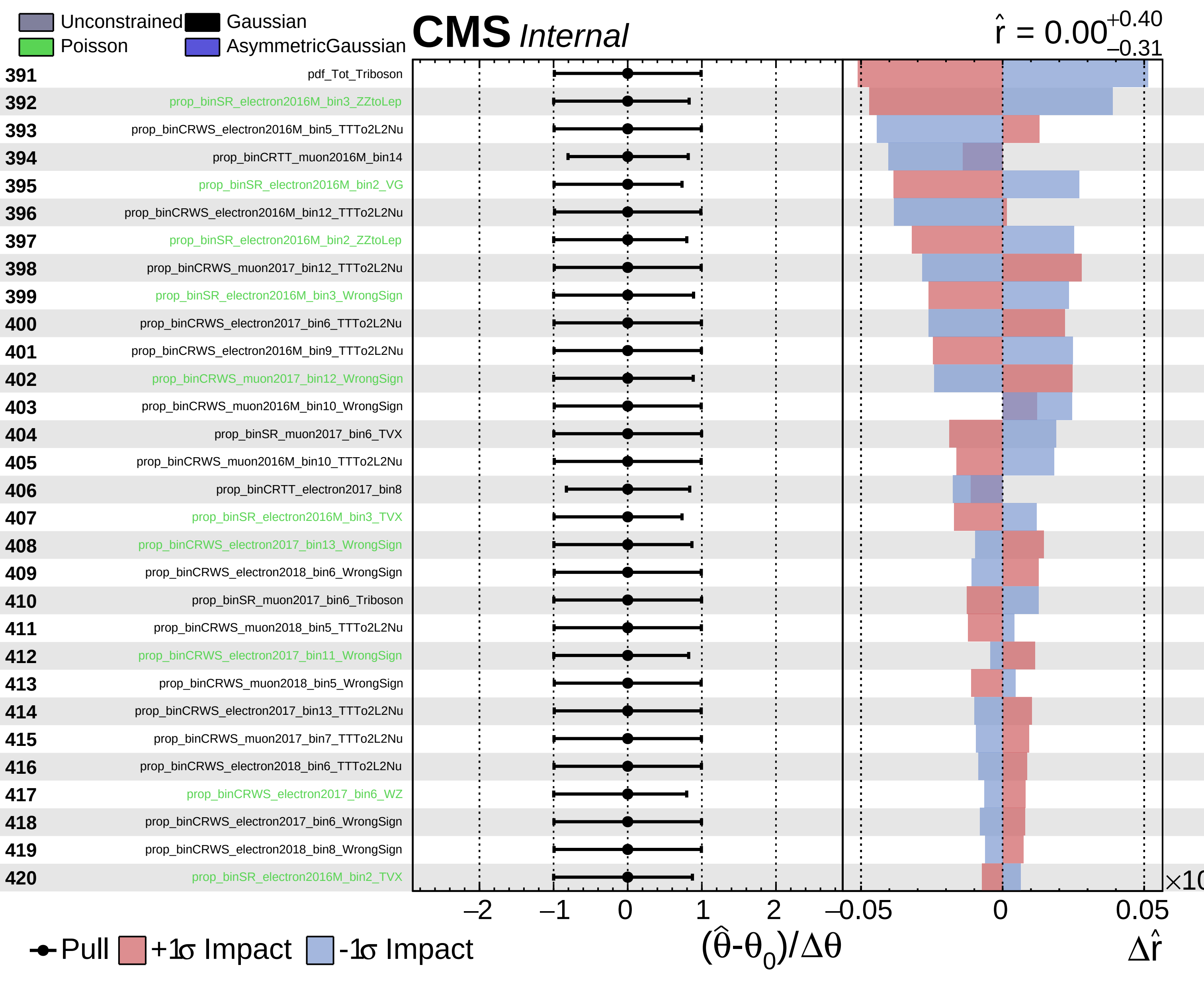


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
301 prop_binCRTT_electron2016M_bin2
302 prop_binCRWS_electron2018_bin12
303 prop_binCRTT_muon2016M_bin12
304 prop_binCRTT_muon2017_bin4
305 prop_binCRTT_muon2017_bin10
306 prop_binCRTT_electron2017_bin10
307 ISR
308 tau_vsjet_2018
309 prop_binCRTT_electron2018_bin1
310 prop_binCRTT_muon2017_bin7
311 prop_binCRWS_muon2017_bin16
312 prop_binCRTT_electron2018_bin10
313 prop_binSR_electron2017_bin0
314 prop_binCRTT_electron2017_bin7
315 prop_binCRTT_electron2017_bin6
316 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
317 prop_binCRTT_electron2016M_bin14
318 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
319 prop_binCRTT_electron2017_bin5
320 prop_binCRTT_electron2017_bin13
321 prop_binCRTT_electron2018_bin11
322 prop_binCRTT_muon2018_bin9
323 prop_binCRTT_electron2018_bin4
324 prop_binCRWS_electron2018_bin8_Fake
325 prop_binCRTT_muon2017_bin16
326 prop_binCRTT_muon2017_bin15
327 prop_binCRTT_electron2016M_bin8
328 prop_binCRTT_muon2017_bin9
329 prop_binCRTT_electron2018_bin15
330 prop_binCRTT_muon2016M_bin11
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

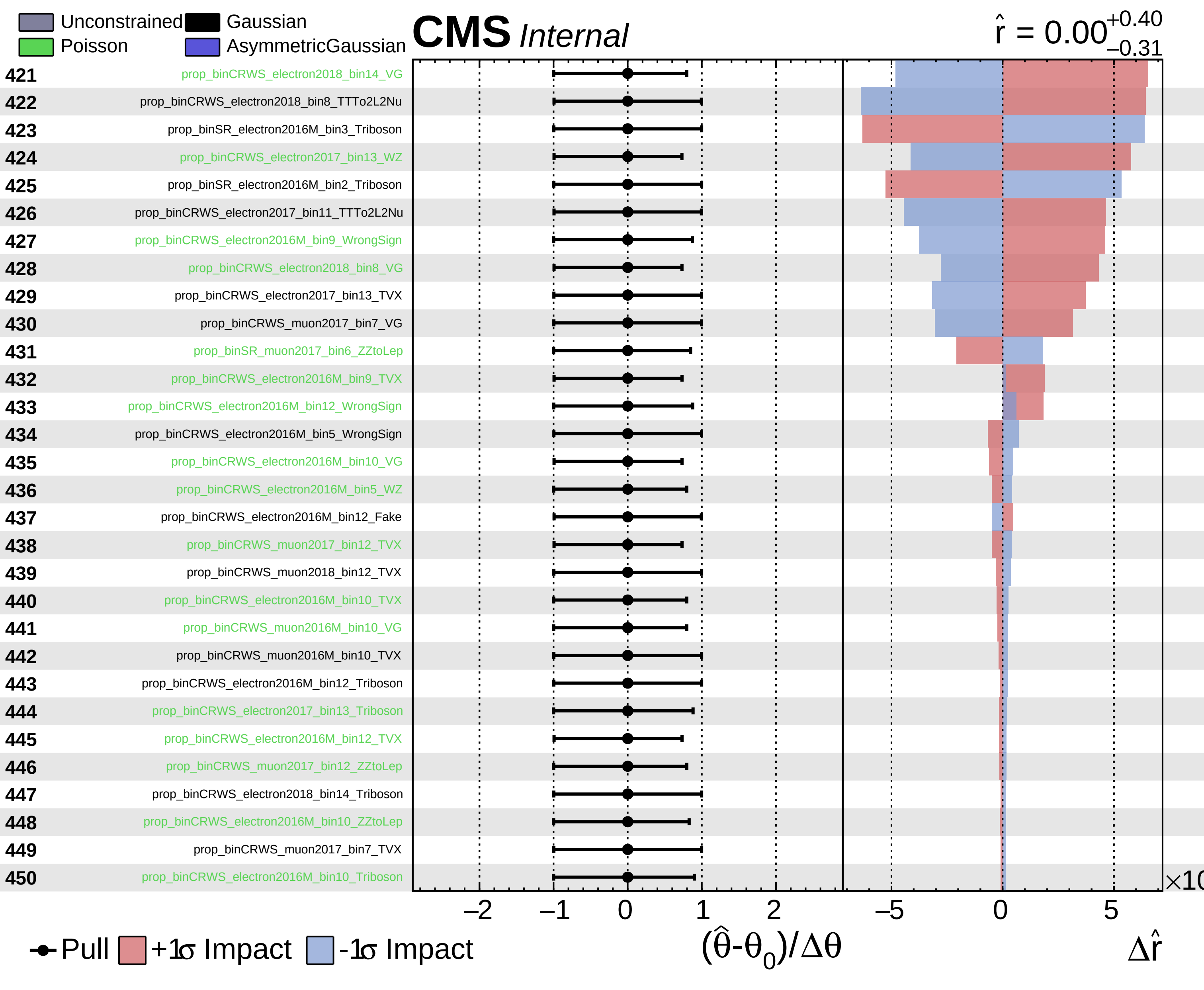


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
331 prop_binCRWS_electron2016M_bin9_Fake
332 prop_binCRTT_muon2016M_bin5
333 prop_binCRTT_muon2018_bin14
334 prop_binCRWS_muon2016M_bin15
335 prop_binCRTT_muon2016M_bin4
336 prop_binCRTT_muon2018_bin7
337 prop_binCRWS_electron2018_bin11
338 prop_binSR_muon2016M_bin0
339 prop_binCRWS_muon2016M_bin10_Fake
340 prop_binCRTT_electron2016M_bin12
341 lep_2018
342 prop_binCRTT_muon2017_bin6
343 prop_binCRTT_muon2016M_bin13
344 prop_binCRWS_electron2017_bin11_Fake
345 prop_binCRWS_muon2016M_bin14
346 prop_binCRTT_electron2017_bin15
347 prop_binCRTT_electron2018_bin13
348 lep_2017
349 prop_binCRTT_muon2017_bin12
350 prop_binCRTT_muon2016M_bin3
351 prop_binCRTT_muon2017_bin2
352 prop_binCRTT_electron2016M_bin9
353 prop_binCRTT_muon2016M_bin8
354 prop_binSR_muon2017_bin6_WrongSign
355 prop_binCRTT_electron2016M_bin7
356 prop_binCRTT_muon2018_bin8
357 prop_binCRTT_muon2016M_bin0
358 prop_binCRTT_electron2016M_bin15
359 prop_binCRTT_electron2018_bin12
360 prop_binCRTT_muon2016M_bin1
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

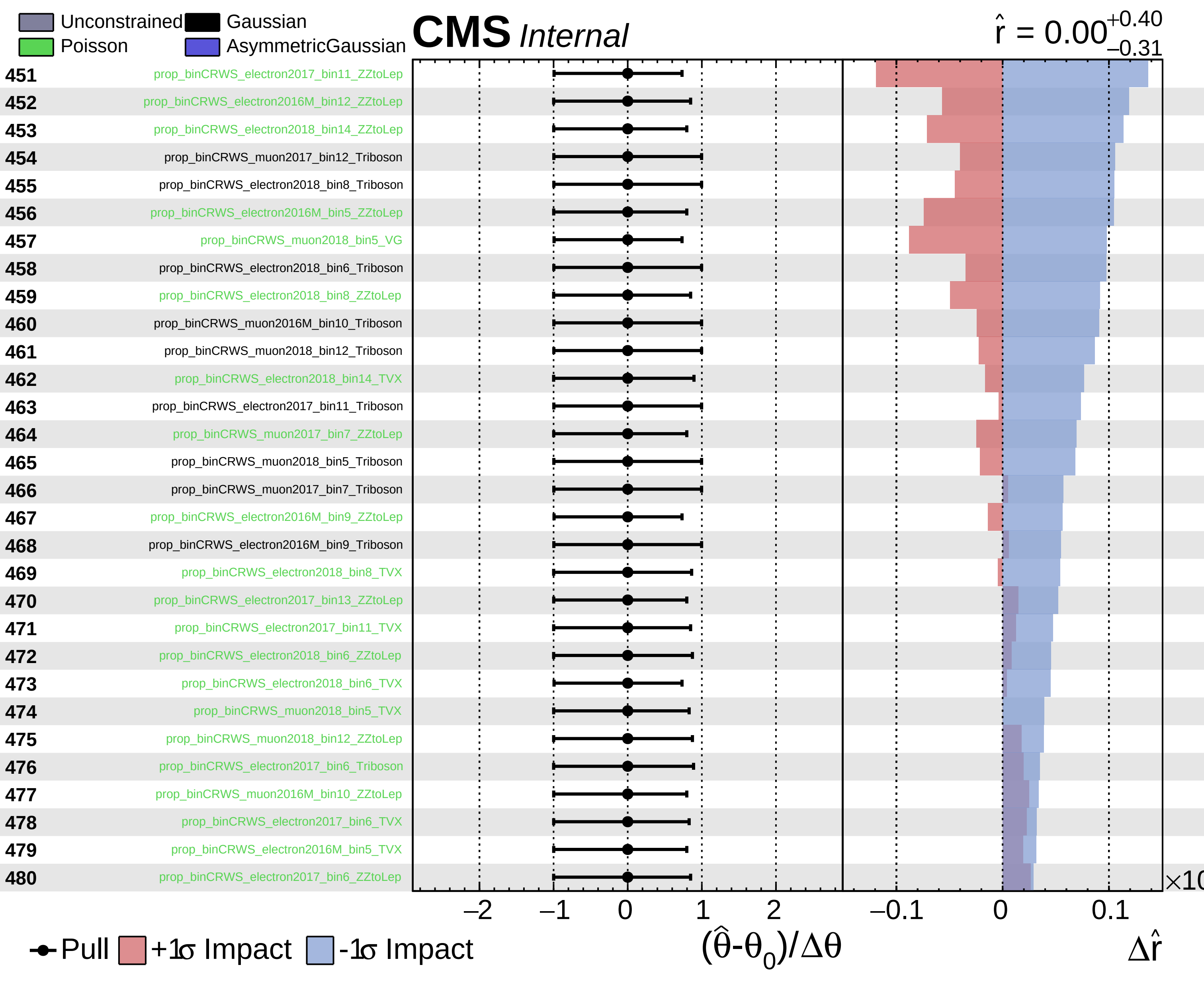

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.40 −0.31
361 prop_binSR_muon2017_bin6_TTTo2L2Nu
362 prop_binSR_electron2016M_bin2_WZ
363 prop_binCRTT_muon2017_bin11
364 prop_binCRTT_electron2018_bin6
365 prop_binCRTT_electron2016M_bin11
366 prop_binCRWS_electron2018_bin14_Fake
367 prop_binCRTT_electron2016M_bin10
368 prop_binCRWS_electron2017_bin13_Fake
369 prop_binCRWS_electron2016M_bin15
370 prop_binSR_electron2016M_bin2_WrongSign
371 prop_binSR_electron2016M_bin3_VG
372 prop_binSR_electron2016M_bin2_TTTo2L2Nu
373 prop_binCRWS_electron2018_bin14_WrongSign
374 prop_binCRWS_muon2018_bin5_Fake
375 pdf_Tot_VG
376 prop_binCRWS_muon2018_bin12_WrongSign
377 prop_binSR_electron2016M_bin3_TTTo2L2Nu
378 prop_binCRTT_electron2017_bin14
379 prop_binCRWS_electron2016M_bin10_WrongSign
380 prop_binCRWS_muon2018_bin12_TTTo2L2Nu
381 prop_binCRTT_muon2016M_bin7
382 prop_binCRTT_electron2016M_bin4
383 prop_binCRWS_electron2017_bin14
384 prop_binCRWS_muon2017_bin12_Fake
385 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
386 prop_binCRWS_muon2017_bin15
387 prop_binCRWS_muon2017_bin7_WrongSign
388 lep_2016M
389 prop_binCRTT_electron2016M_bin5
390 prop_binCRWS_electron2018_bin14_TTTo2L2Nu
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂



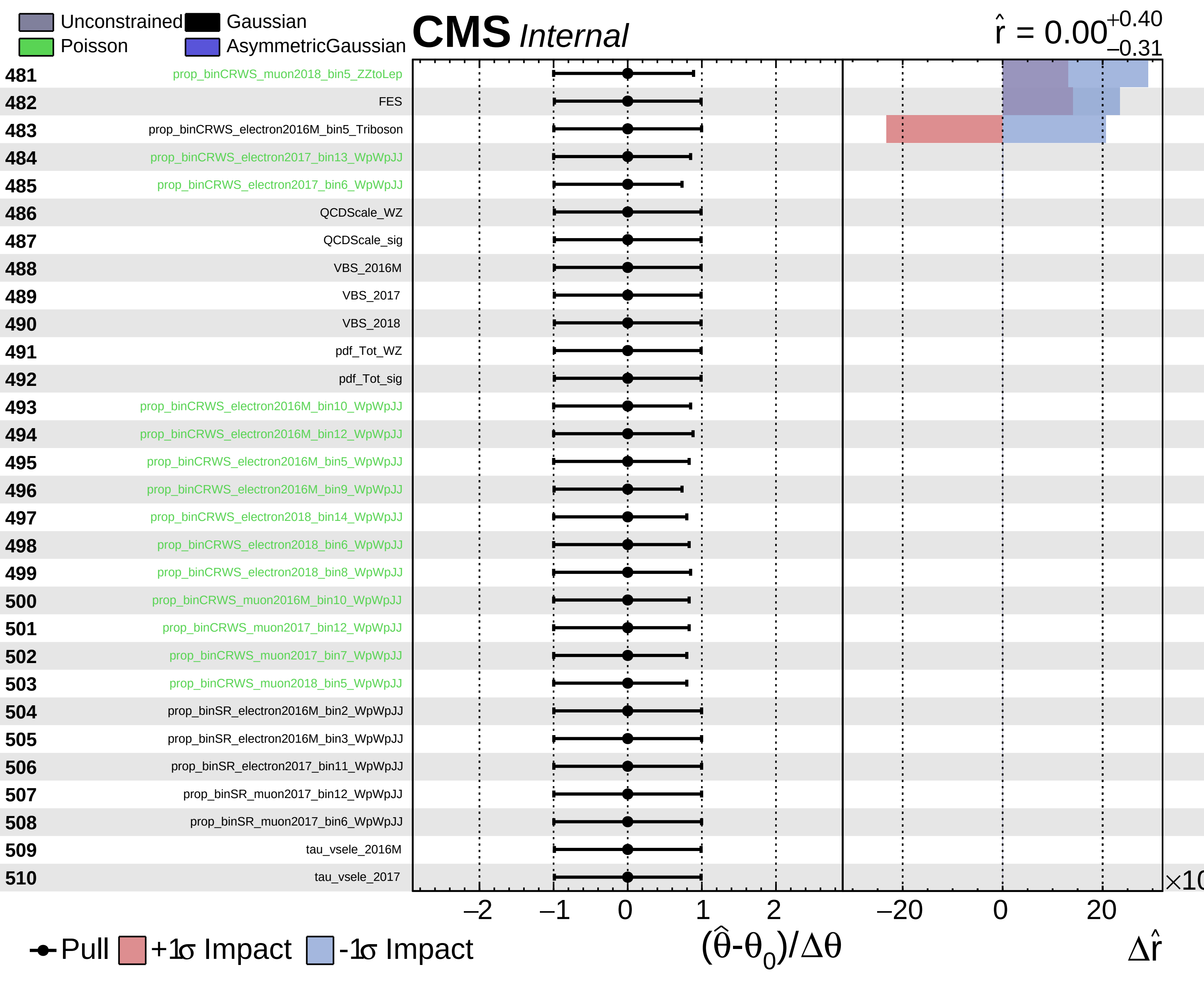
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
391 pdf_Tot_Triboson
392 prop_binSR_electron2016M_bin3_ZZtoLep
393 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
394 prop_binCRTT_muon2016M_bin14
395 prop_binSR_electron2016M_bin2_VG
396 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
397 prop_binSR_electron2016M_bin2_ZZtoLep
398 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
399 prop_binSR_electron2016M_bin3_WrongSign
400 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
401 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
402 prop_binCRWS_muon2017_bin12_WrongSign
403 prop_binCRWS_muon2016M_bin10_WrongSign
404 prop_binSR_muon2017_bin6_TVX
405 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
406 prop_binCRTT_electron2017_bin8
407 prop_binSR_electron2016M_bin3_TVX
408 prop_binCRWS_electron2017_bin13_WrongSign
409 prop_binCRWS_electron2018_bin6_WrongSign
410 prop_binSR_muon2017_bin6_Triboson
411 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
412 prop_binCRWS_electron2017_bin11_WrongSign
413 prop_binCRWS_muon2018_bin5_WrongSign
414 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
415 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
416 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
417 prop_binCRWS_electron2017_bin6_WZ
418 prop_binCRWS_electron2017_bin6_WrongSign
419 prop_binCRWS_electron2018_bin8_WrongSign
420 prop_binSR_electron2016M_bin2_TVX
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



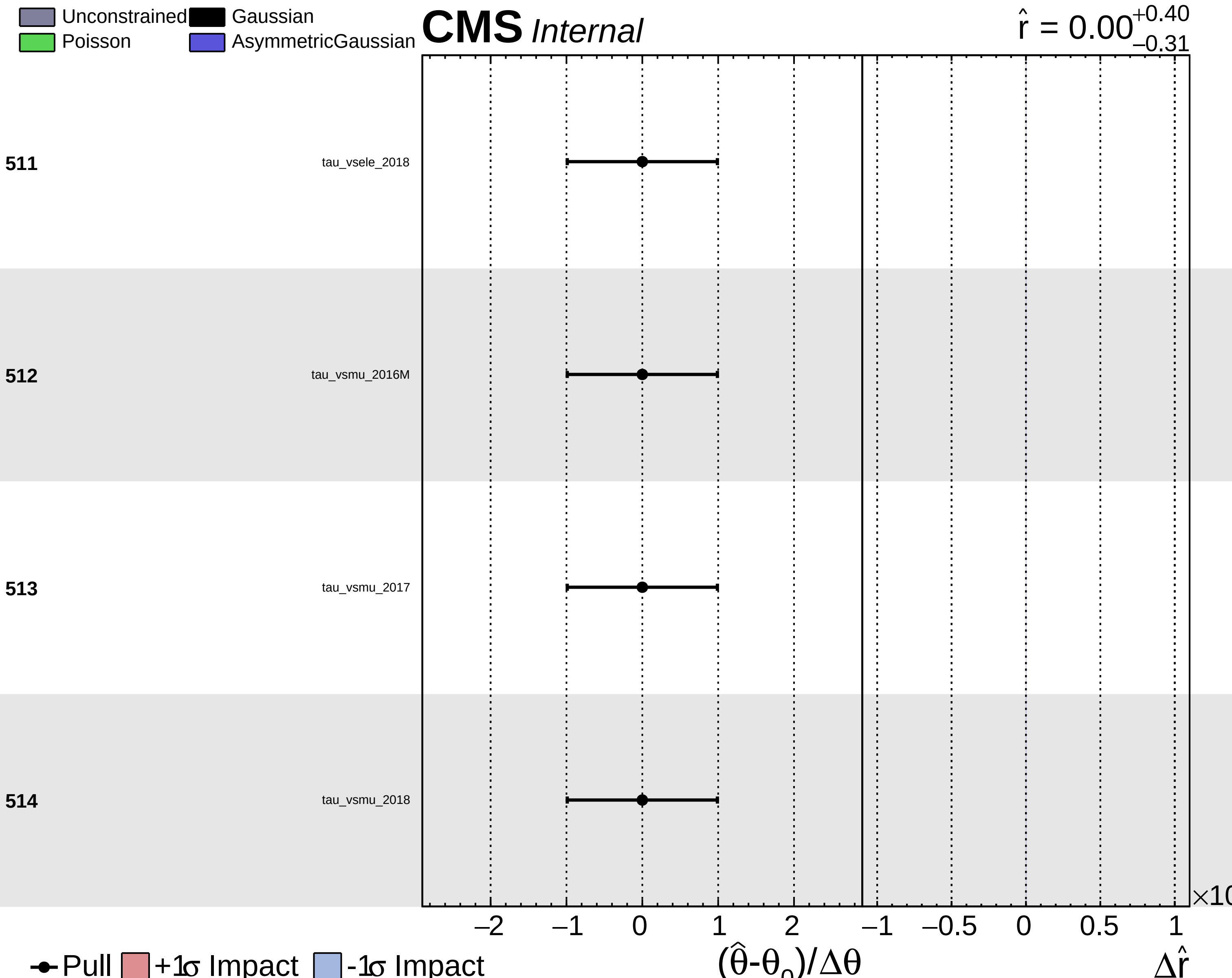
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
421 prop_binCRWS_electron2018_bin14_VG
422 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
423 prop_binSR_electron2016M_bin3_Triboson
424 prop_binCRWS_electron2017_bin13_WZ
425 prop_binSR_electron2016M_bin2_Triboson
426 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
427 prop_binCRWS_electron2016M_bin9_WrongSign
428 prop_binCRWS_electron2018_bin8_VG
429 prop_binCRWS_electron2017_bin13_TVX
430 prop_binCRWS_muon2017_bin7_VG
431 prop_binSR_muon2017_bin6_ZZtoLep
432 prop_binCRWS_electron2016M_bin9_TVX
433 prop_binCRWS_electron2016M_bin12_WrongSign
434 prop_binCRWS_electron2016M_bin5_WrongSign
435 prop_binCRWS_electron2016M_bin10_VG
436 prop_binCRWS_electron2016M_bin5_WZ
437 prop_binCRWS_electron2016M_bin12_Fake
438 prop_binCRWS_muon2017_bin12_TVX
439 prop_binCRWS_muon2018_bin12_TVX
440 prop_binCRWS_electron2016M_bin10_TVX
441 prop_binCRWS_muon2016M_bin10_VG
442 prop_binCRWS_muon2016M_bin10_TVX
443 prop_binCRWS_electron2016M_bin12_Triboson
444 prop_binCRWS_electron2017_bin13_Triboson
445 prop_binCRWS_electron2016M_bin12_TVX
446 prop_binCRWS_muon2017_bin12_ZZtoLep
447 prop_binCRWS_electron2018_bin14_Triboson
448 prop_binCRWS_electron2016M_bin10_ZZtoLep
449 prop_binCRWS_muon2017_bin7_TVX
450 prop_binCRWS_electron2016M_bin10_Triboson
−2
−1
0
1
2
−5
0
5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
451 prop_binCRWS_electron2017_bin11_ZZtoLep
452 prop_binCRWS_electron2016M_bin12_ZZtoLep
453 prop_binCRWS_electron2018_bin14_ZZtoLep
454 prop_binCRWS_muon2017_bin12_Triboson
455 prop_binCRWS_electron2018_bin8_Triboson
456 prop_binCRWS_electron2016M_bin5_ZZtoLep
457 prop_binCRWS_muon2018_bin5_VG
458 prop_binCRWS_electron2018_bin6_Triboson
459 prop_binCRWS_electron2018_bin8_ZZtoLep
460 prop_binCRWS_muon2016M_bin10_Triboson
461 prop_binCRWS_muon2018_bin12_Triboson
462 prop_binCRWS_electron2018_bin14_TVX
463 prop_binCRWS_electron2017_bin11_Triboson
464 prop_binCRWS_muon2017_bin7_ZZtoLep
465 prop_binCRWS_muon2018_bin5_Triboson
466 prop_binCRWS_muon2017_bin7_Triboson
467 prop_binCRWS_electron2016M_bin9_ZZtoLep
468 prop_binCRWS_electron2016M_bin9_Triboson
469 prop_binCRWS_electron2018_bin8_TVX
470 prop_binCRWS_electron2017_bin13_ZZtoLep
471 prop_binCRWS_electron2017_bin11_TVX
472 prop_binCRWS_electron2018_bin6_ZZtoLep
473 prop_binCRWS_electron2018_bin6_TVX
474 prop_binCRWS_muon2018_bin5_TVX
475 prop_binCRWS_muon2018_bin12_ZZtoLep
476 prop_binCRWS_electron2017_bin6_Triboson
477 prop_binCRWS_muon2016M_bin10_ZZtoLep
478 prop_binCRWS_electron2017_bin6_TVX
479 prop_binCRWS_electron2016M_bin5_TVX
480 prop_binCRWS_electron2017_bin6_ZZtoLep
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.40 −0.31
481 prop_binCRWS_muon2018_bin5_ZZtoLep
482 FES
483 prop_binCRWS_electron2016M_bin5_Triboson
484 prop_binCRWS_electron2017_bin13_WpWpJJ
485 prop_binCRWS_electron2017_bin6_WpWpJJ
486 QCDScale_WZ
487 QCDScale_sig
488 VBS_2016M
489 VBS_2017
490 VBS_2018
491 pdf_Tot_WZ
492 pdf_Tot_sig
493 prop_binCRWS_electron2016M_bin10_WpWpJJ
494 prop_binCRWS_electron2016M_bin12_WpWpJJ
495 prop_binCRWS_electron2016M_bin5_WpWpJJ
496 prop_binCRWS_electron2016M_bin9_WpWpJJ
497 prop_binCRWS_electron2018_bin14_WpWpJJ
498 prop_binCRWS_electron2018_bin6_WpWpJJ
499 prop_binCRWS_electron2018_bin8_WpWpJJ
500 prop_binCRWS_muon2016M_bin10_WpWpJJ
501 prop_binCRWS_muon2017_bin12_WpWpJJ
502 prop_binCRWS_muon2017_bin7_WpWpJJ
503 prop_binCRWS_muon2018_bin5_WpWpJJ
504 prop_binSR_electron2016M_bin2_WpWpJJ
505 prop_binSR_electron2016M_bin3_WpWpJJ
506 prop_binSR_electron2017_bin11_WpWpJJ
507 prop_binSR_muon2017_bin12_WpWpJJ
508 prop_binSR_muon2017_bin6_WpWpJJ
509 tau_vsele_2016M
510 tau_vsele_2017
−2 −1 0 1 2
−20 0 20
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.40 −0.31
511
tau_vsele_2018
512
tau_vsmu_2016M
513
tau_vsmu_2017
514
tau_vsmu_2018
−2
−1
0
1
2
−1
−0.5
0
0.5
1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂